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Question 1 (3 marks)

Using the definition of logical equivalence determine if the following statement is true:

$$p \wedge q \equiv \sim (\sim p \vee \sim q).$$

Write a concluding sentence that explains how the work you have done supports your conclusion.

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim p \vee \sim q$
T	T	F	F	T	F
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	F	T

$p \wedge q \iff \sim (\sim p \vee \sim q)$
T
T
T
T

Since $p \wedge q \iff \sim (\sim p \vee \sim q)$ is a tautology we have

$$p \wedge q \equiv \sim p \vee \sim q,$$

and so the statement is true.

Question 2 (4 marks)

Express each of the following as statement in first-order logic.

(a) (a_n) is bounded below and not bounded above.

$$[(\exists r \in \mathbb{R})[(\forall n \in \mathbb{N}) \;\; r < a_n]] \wedge [(\forall s \in \mathbb{R})[(\exists n \in \mathbb{N}) \;\; a_n > s]]$$

(b) Every real number has a multiplicative inverse.

$$(\forall r \in \mathbb{R})[(\exists r' \in \mathbb{R}) \;\; rr' = 1]$$

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Question 3 (5 marks)

Using a proof by induction, prove the following statement for all $n \geq 1$:

$$2^1 + 2^2 + \cdots + 2^n = 2^{n+1} - 2.$$

We proceed by induction on n. Let $p(n)$ be the statement

$$\sum_{t=1}^{n} 2^t = 2^{n+1} - 2$$

When $n = 1$ this statement becomes

$$2^1 = 2^2 - 2,$$

which is true. Therefore $p(1)$ is true.
Assume the statement is true when $n = k$. That is, assume

$$\sum_{t=1}^{k} 2^t = 2^{k+1} - 2$$

When $n = k + 1$ we have

$$\sum_{t=1}^{k+1} 2^t = 2^{k+1} + \sum_{t=1}^{k} 2^t$$

Since $p(k)$ is true, we have $\sum_{t=1}^{k} 2^t = 2^{k+1} - 2$ Therefore

$$\sum_{t=1}^{k+1} 2^t = 2^{k+1} + 2^{k+1} - 2 = 2^{k+2} - 2$$

Therefore

$$\sum_{t=1}^{k+1} 2^t = 2^{k+2} - 2$$

and so $p(k + 1)$ is true.
The result holds by the Principle of Mathematical Induction.

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Question 4 (5 marks)

Let A be a cut. Prove

$$A = \bigcup_{q \in A} (q)_{\mathbf{R}^\star}.$$

Consider $z \in A$. By definition of cut, there exists $z' \in A$ such that $z' > z$. Therefore $z \in (z')_{\mathbf{R}^\star}$ and so $z \in \bigcup_{q \in A} (q)_{\mathbf{R}^\star}$. Therefore $A \subseteq \bigcup_{q \in A} (q)_{\mathbf{R}^\star}$.

Consider $z \in \bigcup_{q \in A} (q)_{\mathbf{R}^\star}$ Since $z \in \bigcup_{q \in A} (q)_{\mathbf{R}^\star}$, there exists $q \in A$ such that $z \in (q)_{\mathbf{R}^\star}$. Therefore $z < q$. Since $q \in A$, the definition of cut implies $z \in A$. Therefore $\bigcup_{q \in A} (q)_{\mathbf{R}^\star} \subseteq A$.

Therefore $A = \bigcup_{q \in A} (q)_{\mathbf{R}^\star}$.

Question 5 (2 marks)

Let A be the cut corresponding to the real number $1 + \sqrt{5}$. Is $\sqrt{2} \in A$? Briefly justify your response.

No, as $\sqrt{2} \notin \mathbb{Q}$ and $A \subset \mathbb{Q}$.

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Question 6 (4 marks)

For each of the following, state whether there exists a sequence that satisfies the given criteria. For those where the sequence exists, give an example. For those where the sequence does not exist, write `there is no such sequence`.

(a) (f_n) does not converge and is bounded above and below by 2 and -2, respectively.

$$f_n = (-1)^n$$

(b) (f_n) is monotone increasing, bounded above and does not converge.

There is no such sequence

(c) (f_n) does not converge and has a convergent subsequence.

$$f_n = (-1)^n$$

(d) (f_n) converges and the subsequence $(f_1, f_3, f_5, \dots)$ diverges.

There is no such sequence.

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Question 7 (5 marks)

Let $L \in \mathbb{R}$. Let (f_n) be a sequence such that

$$(f_1, f_3, f_5, \dots) \to L \qquad \text{and} \qquad (f_2, f_4, f_6, \dots) \to L.$$

Prove $f_n \to L$.

Let $(g_n) = (f_1, f_3, f_5, \dots)$ and let $(h_n) = (f_2, f_4, f_6, \dots)$. Let $\epsilon > 0$. By hypothesis, there exists M_g and M_h such that

- $|g_k - L| < \epsilon$ whenever $k > M_g$; and
- $|h_k - L| < \epsilon$ whenever $k > M_h$.

Let $M = \max\{M_g, M_h\}$ and consider $n > 2M$. We proceed based on the parity of n.

Notice that when n is even we have $f_n = h_{n/2}$. Since $n > 2M$ we have $n/2 > M \geq M_h$. Therefore

$$|f_n - L| = |h_{n/2} - L| < \epsilon$$

Similarly when n is odd, we have $f_n = g_{(n+1)/2}$. Since $n > 2M$ we have $(n+1)/2 > M \geq M_g$. Therefore

$$|f_n - L| = |g_{(n+1)/2} - L| < \epsilon$$

Therefore for every $\epsilon > 0$ there exists $M \in \mathbb{N}^+$ such that $|f_n - L| < \epsilon$ whenever $n > M$. Therefore $f_n \to L$.

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Question 8 (5 marks)

Using an ϵ-δ proof, prove

$$\lim_{x \to 4} \frac{x^2}{x-3} = 16.$$

Use this page for any scratch work. Write your proof on the subsequent page. Hand in this page as part of your GradeScope submission as work you do on this page may be used to award partial credit. In your proof you may refer back to any computations you do on this page. In doing so, please make clear where to find the relevant computation on this page.

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Let $\epsilon > 0$. Let $\delta = \min\{1/2, \epsilon/17\}$.
Notice that when x satisfies $0 < |x-4| < \frac{1}{2}$ we have

$$\frac{1}{|x-3|} < 2 \text{ and } |x-12| < \frac{17}{2}$$

When $0 < |x-4| < \delta$,

$$\begin{aligned}\left|\frac{x^2}{x-3} - 16\right| &= \left|\frac{(x-12)(x-4)}{x-3}\right| \\ &= |x-12| \cdot \frac{1}{|x-3|} \cdot |x-4| \\ &< 17\delta \\ &\leq \epsilon\end{aligned}$$

Therefore

$$\lim_{x\to 4} \frac{x^2}{x-3} = 16.$$

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Question 9 (5 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ such that

$$\lim_{x\to 0^-} f(x) < \lim_{x\to 0^+} f(x).$$

Prove there exists $\delta > 0$ such that $f(x_1) < f(x_2)$ whenever $-\delta < x_1 < 0$ and $0 < x_2 < \delta$.

Let $\lim_{x\to 0^-} f(x) = L^-$ and $\lim_{x\to 0^+} f(x) = L^+$. Let $\epsilon = \frac{L^+ - L^-}{2}$.
By hypothesis there exists δ^- and δ^+ such that

- $|f(x_2) - L^+| < \epsilon$ whenever $0 < x < \delta^+$ and
- $|f(x_1) - L^-| < \epsilon$ whenever $-\delta^- < x < 0$.

Let $\delta = \min\{\delta^+, \delta^-\}$ consider x_1, x_2 such that $-\delta < x_1 < 0$ and $0 < x_2 < \delta$.
Since $0 < x_2 < \delta \leq \delta^+$, we have

$$L^+ - \epsilon < f(x_2) < L^+ + \epsilon$$

Similarly, for x_1 we have

$$L^- - \epsilon < f(x_1) < L^- + \epsilon$$

Therefore

$$f(x_2) - f(x_1) > L^+ - \epsilon - L^- - \epsilon = L^+ - L^- - 2\epsilon = 0.$$

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Question 10 (5 marks)

Let $g : [-1, 1] \to \mathbb{R}$ such that g is continuous and $0 < g(x) < 1$ for all $x \in [-1, 1]$. Let $(f_n(x))$ be a sequence of functions that converges uniformly to g on $[-1, 1]$. Prove there exists $M \in \mathbb{N}^+$ such that for all $x \in [-1, 1]$ and all $n > M$ the following inequality holds:

$$0 < f_n(x) < 1.$$

By the Extreme Value Theorem, g attains a minimum and a maximum on $[-1, 1]$. Let c and C respectively be these values.

Consider a fixed $\epsilon > 0$ such that $\epsilon < c$ and $\epsilon < 1 - C$. Since $(f_n(x))$ converges uniformly to g on $[-1, 1]$, there exists $M \in \mathbb{N}^+$ such that for all $x \in [-1, 1]$ we have $|f_n(x) - g(x)| < \epsilon$ whenever $n > M$. Unpacking the absolute value, we have

$$g(x) - \epsilon < f_n(x) < g(x) + \epsilon$$

Since $c \leq g(x) \leq C$ we have

$$c - \epsilon < f_n(x) < \epsilon + C$$

By construction, $\epsilon < 1 - C$ and $-c < -\epsilon$. Therefore

$$0 = c - c < c - \epsilon < f_n(x) < \epsilon + C < 1 - C + C = 1$$

and so

$$0 < f_n(x) < 1$$

for all $x \in [-1, 1]$ whenever $n > M$.

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Question 11 (3 marks)

Let

$$f(x) = \begin{cases} x^2 & x > 0 \\ x - 1 & x \leq 0 \end{cases}$$

Using an ϵ-δ argument, prove $f(x)$ is not continuous at 0.

Let $\epsilon = \frac{1}{2}$. For $x > 0$ we have

$$|f(x) - f(0)| = |x^2 + 1| > 1$$

For every $\delta > 0$ there exists $x \in (0 - \delta, 0 + \delta)$ such that $x > 0$.

Therefore for every $\delta > 0$ there exists $x \in (0 - \delta, 0 + \delta)$ such that $|f(x) - f(0)| > 1$.

Therefore for every $\delta > 0$ there exists $x \in (0 - \delta, 0 + \delta)$ such that $|f(x) - f(0)| > \epsilon$.

Therefore $f(x)$ is not continuous at 0.

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Question 12 (5 marks)

Let $f, g : \mathbb{R} \to \mathbb{R}$ be differentiable on $\mathbb{R}$ such that $f'(x) = g'(x)$ for all $x \in \mathbb{R}$. Prove there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in \mathbb{R}$.

Let $h(x) = f(x) - g(x)$.
We prove $h(x)$ is constant by proving $h(b) = h(0)$ for all $b \neq 0$. Without loss of generality, assume $b > 0$.

Note that by linearity of the derivative, h is differentiable on $[0, b]$. Since f and g are differentiable on $\mathbb{R}$, then they are both continuous on $\mathbb{R}$. Hence, by the Algebra of Continuous functions theorem, h is continuous on $[0, b]$.
By the Mean Value Theorem, there exists $c \in (a, b)$ such that

$$0 = f'(c) - g'(c) = h'(c) = \frac{(f(b) - g(b)) - (f(0) - g(0))}{b - a}$$

Therefore

$$h(b) = f(b) - g(b) = f(0) - g(0) = h(0)$$

Therefore for all $x \in \mathbb{R}$ we have

$$f(x) - g(x) = h(0)$$

Therefore

$$f(x) = g(x) + h(0)$$

And so we conclude there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$.

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Question 13 (3 marks)

Consider the following integrable function

$$f(x) = \begin{cases} 2x & x < 2 \\ x^2 & x \geq 2. \end{cases}$$

Find a lower bound for $\int_1^4 f(x)\,dx$ using the partition $P = \{1, 3, 4\}$.

We have

$$L(f, P) \leq \int_0^4 f(x)\,dx$$

We compute

$$L(f, P) = m_1(3 - 1) + m_2(4 - 3)$$

We have $m_1 = 2$ and $m_2 = 9$. And so

$$L(f, P) = 2(2) + 9(1) = 13.$$

Therefore

$$13 \leq \int_1^4 f(x)\,dx$$

Question 14 (3 marks)

Let $f : [x_0, x_n] \to \mathbb{R}$ and let $P = \{x_0, x_1, x_2, \ldots, x_n\}$ be a partition of $[x_0, x_n]$. Prove $L(f, P) \leq U(f, P)$.

Recall $m_k = \inf\{f(x) \mid x \in [x_{k-1}, x_k]\}$ and $M_k = \inf\{f(x) \mid x \in [x_{k-1}, x_k]\}$. Therefore for all $k \in \{1, 2, \ldots n\}$ we have

$$m_k(x_k - x_{k-1}) \leq M_k(x_k - x_{k-1}).$$

Therefore

$$L(f, P) = \sum_{k=1}^{n} m_k(x_k - x_{k-1}) \leq M_k(x_k - x_{k-1}) = U(f, P).$$

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Question 15 (5 marks)

Let

$$f(x) = \begin{cases} 3 & x \in (0,1) \\ 0 & x = 0, 1. \end{cases}$$

Prove $f(x)$ is integrable on $[0,1]$.

Let $\epsilon > 0$. Let $\epsilon_1 = \epsilon_2 = \min\{\frac{1}{4}, \frac{\epsilon}{8}\}$. Consider the partition $P = \{0,\ \epsilon_1,\ 1-\epsilon_2,\ 1\}$. Notice

$$U(f,P) - L(f,P) = 3(\epsilon_1 + \epsilon_2) \leq 3(\tfrac{\epsilon}{8} + \tfrac{\epsilon}{8}) = \tfrac{3\epsilon}{4} < \epsilon.$$

Therefore for every $\epsilon > 0$ there exists a partition P such that $|U(f,P) - L(f,P)| < \epsilon$. Therefore (by the Theorem from Assignment 5) f is integrable.

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Question 16 (4 marks)

Using appropriate tests, determine the convergence or divergence of the following series.

(a)

$$\sum_{n=1}^{\infty}(-1)^n \arctan(n)$$

Since $(-1)^n \arctan(n) \not\to 0$, the series diverges.

(b)

$$\sum_{n=2}^{\infty}\frac{(-1)^n}{2\log n}$$

We apply the alternating series test. Since $\left(\frac{1}{2\log n}\right)$ is monotone decreasing and

$$\left(\frac{1}{2\log n}\right) \to 0$$

the series $\sum_{n=2}^{\infty}\frac{(-1)^n}{2\log n}$ converges.

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Question 17 (5 marks)

Find the interval of convergence of the following power series

$$\sum_{n=1}^{\infty} 3^n (x-5)^n.$$

Notice

$$\frac{3^{n+1}(x-5)^{n+1}}{3^n(x-5)^n} = 3(x-5)$$

Therefore for any $x \in \mathbb{R}$, the resulting series is geometric with ratio $r = 3(x-5)$. A geometric series converges if and only if $|r| < 1$. Simplifying the inequality

$$|3(x-5)| < 1$$

we find $5 - \frac{1}{3} < x < 5 + \frac{1}{3}$.
And so the interval of convergence is $\left(5 - \frac{1}{3}, 5 + \frac{1}{3}\right)$.

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Question 18 (5 marks)

Find the coefficient of x^2 in the order 2 Taylor polynomial of $\sqrt{x+1}+x^2$ at 3.

Let $f(x) = \sqrt{x+1}+x^2$. We have

- $f^{(1)}(x) = \frac{1}{2}(x+1)^{-1/2} + 2x$, and
- $f^{(2)}(x) = -\frac{1}{4}(x+1)^{-3/2} + 2$.

In the order 2 Taylor polynomial of $\sqrt{x+1}+x^2$ at 3 the only x^2 contribution will come from the term: $\frac{f^{(2)}(3)}{2}(x-3)^2$.

We find:

$$\frac{f^{(2)}(3)}{2}(x-3)^2 = \frac{63}{64}(x-3)^2.$$

Therefore the coefficient of x^2 in the in the order 2 Taylor polynomial of $\sqrt{x+1}+x^2$ at 3 is $\frac{63}{64}$.

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Question 19 (3 marks)

Let n be a non-negative integer. Let $C_n(\theta)$ denote the order n Taylor polynomial of $\cos(\theta)$ at 0. Prove

$$|C_n(\alpha) - \cos(\alpha)| < \frac{7^{n+1}}{(n+1)!}$$

for all $\alpha \in (0, 2\pi)$.

Let $\alpha \in (0, 2\pi)$. Let $n \in \mathbb{N}$. By Taylor's Theorem there exists $c \in (0, \alpha)$ where

$$\cos(\alpha) - C_n(\alpha) = \frac{f^{n+1}(c)}{(n+1)!}\alpha^{n+1}$$

We have

$$f^{n+1}(c) \in \{\sin(c), \cos(c), -\sin(c), -\cos(c)\}$$

Therefore

$$-1 \leq f^{n+1}(c) \leq 1$$

Therefore

$$-\frac{\alpha^{n+1}}{(n+1)!} \leq \cos(\alpha) - C_n(\alpha) \leq \frac{\alpha^{n+1}}{(n+1)!}$$

and so

$$|C_n(\alpha) - \cos(\alpha)| \leq \frac{\alpha^{n+1}}{(n+1)!}$$

Since $\alpha \in (0, 2\pi)$ and $2\pi < 7$, then $\alpha < 7$. Therefore

$$|C_n(\alpha) - \cos(\alpha)| \leq \frac{\alpha^{n+1}}{(n+1)!} < \frac{7^{n+1}}{(n+1)!}$$

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Question 20 (3 marks)

Throughout our time in MAST20026 we spent a lot of time thinking about *how* mathematics is done in addition to actually doing mathematics. How did the ideas of this subject enlarge your sense of what it means to *do* mathematics? In your response, cite two examples of things you saw in MAST20026 that enlarged your sense of what it means to do mathematics.

Students will be given 3/3 if they write something reasonable and cite two examples. They will get 2/3 for something reasonable with 1 example and 1/3 for writing something reasonable.
These three marks are essentially *free* and balance out the difficulty of the rest of the exam.

Turn the page for appended material.

Do not submit appended material with exam submission.

End of Exam — Total Available Marks = 82